Note B: baseline and robustness results

Unemployment and prices after a monetary policy surprise

Generated by `baseline/06_report.py`

Specification

Each number is the response of the unemployment gap (percentage points) or the core PCE price *level* (percent) to a monetary policy surprise that raises the 2-year Treasury yield by 1 percentage point in the announcement window, 12, 24 and 36 months later. Responses are traced through a daily block (the 2-year yield and, where included, the market financial conditions index (FCI) of Note A) into a monthly system for the unemployment gap and core PCE inflation. **Main tables use 12 monthly lags, fixed in advance; the appendix shows 3, 6 and 12.** The surprises are the Nakamura–Steinsson policy news series (252 FOMC announcements, January 1996 to July 2026), identified in the announcement window with macro variables held fixed. Macro variables come first within the month. No deterministic trend is included except in Robustness 2c–d. The sample runs to September 2026 unless stated. Intervals are 90% bootstrap intervals from 200 draws (resampling announcements and monthly residuals; the daily block is not re-estimated). A star marks a 90% interval that excludes zero. Without the FCI the daily block contains the 2-year yield alone; with it, the FCI is in the same form (nominal, real or r*-gap) as the 2-year yield.

1 Baseline

Table 1: Baseline: 1996 to the present, all data in, no trend (12 lags)

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
Nominal	none	12	369	−0.84*	−0.46*	−0.25	0.09	0.01	−0.13
				[−1.27, −0.39]	[−0.76, −0.13]	[−0.60, 0.10]	[−0.11, 0.34]	[−0.37, 0.45]	[−0.70, 0.51]
	nominal	12	369	−0.01	0.30	0.38*	−0.15	−0.49*	−0.82*
				[−0.54, 0.45]	[−0.05, 0.60]	[0.05, 0.69]	[−0.41, 0.10]	[−0.94, −0.01]	[−1.43, −0.12]
Real	none	12	369	−0.62*	−0.33	−0.20	−0.04	−0.11	−0.22
				[−0.96, −0.19]	[−0.64, 0.03]	[−0.52, 0.15]	[−0.24, 0.17]	[−0.46, 0.22]	[−0.76, 0.27]
	real	12	369	0.13	0.33	0.31*	−0.25*	−0.53*	−0.73*
				[−0.30, 0.55]	[−0.03, 0.67]	[0.01, 0.63]	[−0.47, −0.03]	[−0.97, −0.07]	[−1.43, −0.07]
r*-gap	none	12	369	−0.28	−0.06	0.04	−0.22*	−0.44*	−0.68*
				[−0.91, 0.18]	[−0.68, 0.31]	[−0.56, 0.47]	[−0.40, −0.01]	[−0.82, −0.03]	[−1.34, −0.07]
	r*-gap	12	369	0.33	0.50*	0.43*	−0.41*	−0.78*	−1.06*
				[−0.17, 0.67]	[0.12, 0.76]	[0.10, 0.70]	[−0.59, −0.19]	[−1.25, −0.35]	[−1.90, −0.32]

Nominal, real and r*-gap 2-year yield; the FCI is Note A's index in the matching form in each case. Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.



Figure 1: Baseline: the three steps, without and with the FCI

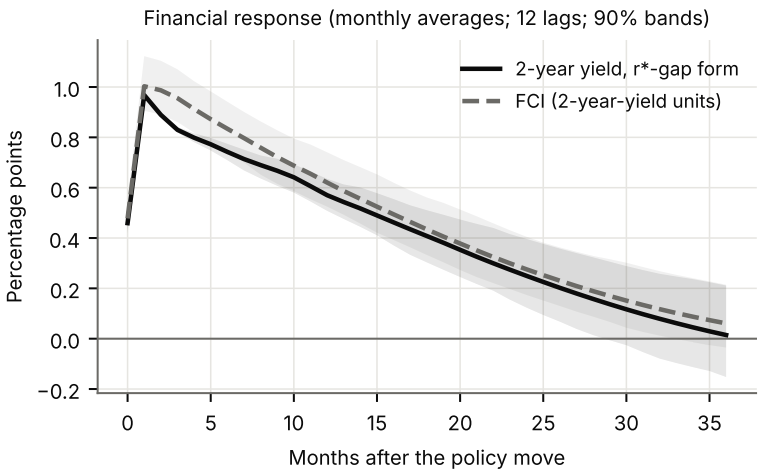


Figure 2: The financial response behind the baseline

Table 2: Market-index weights from policy news

Component	Response to policy news	Weight	Share of a policy move
10-year Treasury yield (pp)	+0.53 (0.06)	+0.609	0.33
Equity valuation, 100 ln CAPE (%)	−7.78 (0.93)	−0.051	0.40
Real broad dollar, 100 ln (%)	+4.22 (0.53)	+0.065	0.28

Response of each 30-minute announcement-window change to the policy-news factor (scaled so the 2-year yield moves 1 pp), with robust standard errors; weights from the one-factor GLS of Note A; share = weight × response (the three shares sum to 1).

Table 3: Implied trade-off: Phillips multiplier κ_H (12 lags)

Case	FCI	$H = 24$ months	$H = 36$ months
Nominal	none	−0.01 [−0.31, 0.42]	0.08 [−0.25, 0.66]
	nominal	unbounded	unbounded
Real	none	0.14 [−0.24, 1.37]	0.21 [−0.22, 3.19]
	real	unbounded	unbounded
r*-gap	none	unbounded	unbounded
	r*-gap	−1.06 [−6.84, −0.40]	−0.88 [−3.34, −0.27]
Start 1973	r*-gap	−2.16 [−27.06, −1.20]	−2.19 [−29.88, −1.29]
2020 dropped	r*-gap	−0.95 [−1.83, −0.38]	−0.80 [−1.74, −0.27]
2020 dropped, start 1973	r*-gap	−1.47 [−2.50, −0.89]	−1.59 [−2.84, −0.90]
Ends 2019	r*-gap	−0.35 [−1.23, 0.15]	−0.38 [−2.98, 0.12]
Ends 2019, start 1973	r*-gap	−1.03 [−1.87, −0.48]	−1.25 [−2.42, −0.67]
r*-gap minus nominal, with FCI		[−1.57, 6.37]	[−0.89, 4.05]

κ_H = core PCE price level at month H (percent) per percentage-point year of cumulative unemployment-gap response (Barnichon and Mesters 2021); the sacrifice ratio is $-1/\kappa_H$. Point estimate and 90% confidence set (Anderson–Rubin type) in brackets, computed exactly (a wide grid refined by bisection), so a finite end is never an artifact of the grid. An infinite end means that the 90% interval of the cumulative unemployment response includes zero: the data then cannot rule out arbitrarily steep trade-offs on that side; "unbounded": both sides. Last row: paired difference over common announcement draws.

1.1 Monthly VAR benchmark

The baseline traces a surprise through a daily block. The benchmark in most of the literature is a purely monthly VAR. We estimate it on the same announcements, for the baseline sample (from 1996) and for a long sample (from 1973),

$$y_t = c + \sum_{l=1}^{L} A_l\, y_{t-l} + u_t, \qquad y_t = (\text{unemployment gap},\ \text{core PCE inflation},\ i_t\,[,\ F_t])', \tag{1}$$

with i_t the monthly-average 2-year yield and F_t the FCI in the same step. Responses to a unit shock with impact vector s are $\Theta_h s$, where Θ_h are the moving-average matrices of (1). Two ways of choosing s:

- **Proxy VAR** (Stock and Watson 2018; Mertens and Ravn 2013; Gertler and Karadi 2015; with high-frequency surprises, Bauer and Swanson 2023). The instrument z_t is the Nakamura–Steinsson surprise (Nakamura and Steinsson 2018) summed over the announcements in month t. Assumptions: relevance, $E[z_t \varepsilon_t^{mp}] \neq 0$, and exogeneity, $E[z_t \varepsilon_t^{o}] = 0$ for the other shocks. Then $s_j/s_i = E[z_t u_{jt}]/E[z_t u_{it}]$, estimated by regressing each residual on z_t; every entry of s, including unemployment and inflation, is estimated. Inference: the 90% Anderson–Rubin set $\{\theta :\ \hat{b}(\theta)^2/\hat{V}(\theta) \leq \chi^2_{1,0.90}\}$ for the response of a target (conditional on the VAR coefficients, HC1

variance), a quadratic inequality in θ solved exactly (Montiel Olea, Stock and Watson 2021; Lunsford 2015).

- **Window impact.** $s = \phi\,(0, 0, 1, b_F)'$: the impact on the 2-year yield and the FCI is measured in the announcement window, the macro entries are zero, and ϕ is the average share of the month that follows the announcement. The assumption, as in the main model, is that the shock reaches unemployment and inflation only through the 2-year yield and the FCI. Inference: bootstrap.

The two units differ (monthly-average versus window 2-year). Plagborg-Møller and Wolf (2021) show that VARs and local projections estimate the same responses; the difference here is the temporal aggregation of a surprise that is concentrated on one day.

Table 4: Monthly VAR benchmark, 1996 sample (12 lags)

					Unemployment gap (pp)			Price level (%)		
2-year	FCI	Identification	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
Nominal	none	Proxy VAR	12	369	−1.27*	−0.84*	−0.45	0.07	−0.07	−0.32
					[−1.93, −0.68]	[−1.42, −0.25]	[−0.92, 0.06]	[−1.31, 1.25]	[−2.05, 1.59]	[−2.58, 1.56]
		Window impact	12	369	−0.57*	−0.39*	−0.21	0.07	0.02	−0.08
					[−0.88, −0.33]	[−0.74, −0.17]	[−0.57, 0.05]	[−0.07, 0.21]	[−0.26, 0.33]	[−0.48, 0.33]
	nominal	Proxy VAR	12	369	−0.87	−0.80	−0.51	−0.36	−0.47	−0.52
					unbounded	unbounded	unbounded	unbounded	unbounded	unbounded
		Window impact	12	369	−0.16	−0.10	−0.05	−0.05	−0.20	−0.33
					[−0.47, 0.12]	[−0.41, 0.23]	[−0.39, 0.27]	[−0.20, 0.08]	[−0.43, 0.06]	[−0.65, 0.09]
Real	none	Proxy VAR	12	369	−0.89*	−0.49	−0.16	−0.31	−0.58	−0.83
					[−1.77, −0.31]	[−1.23, 0.24]	[−0.74, 0.54]	[−2.00, 0.87]	[−3.03, 1.05]	[−3.68, 0.97]
		Window impact	12	369	−0.39*	−0.25*	−0.12	−0.02	−0.08	−0.15
					[−0.69, −0.21]	[−0.60, −0.02]	[−0.45, 0.16]	[−0.14, 0.10]	[−0.32, 0.15]	[−0.56, 0.21]
	real	Proxy VAR	12	369	−0.38	−0.46	−0.34	−0.72	−0.87	−0.82
					unbounded	unbounded	unbounded	unbounded	unbounded	unbounded
		Window impact	12	369	−0.02	0.00	0.01	−0.13	−0.26	−0.33
					[−0.32, 0.26]	[−0.32, 0.27]	[−0.33, 0.35]	[−0.32, 0.04]	[−0.60, 0.06]	[−0.78, 0.08]
r*-gap	none	Proxy VAR	12	369	−0.47	−0.07	0.15	−0.51	−0.89	−1.20
					[−1.52, 0.08]	[−0.86, 0.52]	[−0.37, 0.63]	[−1.90, 0.56]	[−2.88, 0.55]	[−3.46, 0.34]
		Window impact	12	369	−0.15	−0.02	0.05	−0.12*	−0.24*	−0.34*
					[−0.42, 0.11]	[−0.26, 0.22]	[−0.14, 0.29]	[−0.25, −0.01]	[−0.49, −0.01]	[−0.68, −0.02]
	r*-gap	Proxy VAR	12	369	−0.06	−0.06	−0.10	−0.89	−1.30	−1.43
					[−1.82, 5.22]	[−1.57, 3.84]	[−1.18, 2.00]	[−6.80, 0.28]	[−9.27, 0.41]	[−9.64, 0.47]
		Window impact	12	369	0.15	0.16	0.12	−0.21*	−0.37*	−0.45*
					[−0.14, 0.38]	[−0.10, 0.40]	[−0.12, 0.43]	[−0.35, −0.07]	[−0.64, −0.12]	[−0.84, −0.13]

Monthly VAR in the unemployment gap, core PCE inflation, the 2-year yield and, when included, the FCI (same step), sample as in the title, to September 2026, no trend, nothing dropped. Proxy VAR: Nakamura–Steinsson surprise as instrument; sets are exact 90% Anderson–Rubin sets, "unbounded" when the instrument cannot reject large effects; unit +1 pp on the monthly-average 2-year yield. Window impact: impact vector measured in the announcement window and scaled by the share of the month after the announcement; 90% bootstrap intervals; unit +1 pp 2-year in the window. The two units differ. * = 90% interval excludes zero.

Table 5: Monthly VAR benchmark, 1973 sample (12 lags)

					Unemployment gap (pp)			Price level (%)		
2-year	FCI	Identification	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
Nominal	none	Proxy VAR	12	633	−0.57*	−0.30*	−0.11	0.05	0.09	0.09
					[−1.31, −0.16]	[−0.69, −0.09]	[−0.34, 0.03]	[−1.69, 1.59]	[−2.76, 2.62]	[−3.69, 3.45]
		Window impact	12	633	−0.16*	−0.08	−0.02	0.05	0.09	0.12
					[−0.26, −0.05]	[−0.18, 0.00]	[−0.11, 0.07]	[−0.04, 0.14]	[−0.13, 0.27]	[−0.24, 0.40]
	nominal	Proxy VAR	12	633	0.33	0.21	0.15	−0.50	−0.93	−1.34
					unbounded	unbounded	unbounded	unbounded	unbounded	unbounded
		Window impact	12	633	0.15*	0.11*	0.06	0.04	0.04	0.06
					[0.01, 0.34]	[0.00, 0.30]	[−0.02, 0.21]	[−0.08, 0.16]	[−0.22, 0.29]	[−0.33, 0.40]
Real	none	Proxy VAR	12	633	−0.56*	−0.30*	−0.14	−0.25	−0.51	−0.75
					[−2.14, −0.11]	[−1.14, −0.06]	[−0.61, 0.02]	[−2.58, 1.81]	[−4.36, 2.88]	[−5.84, 3.70]
		Window impact	12	633	−0.10	−0.05	−0.02	−0.10*	−0.21*	−0.30*
					[−0.21, 0.00]	[−0.16, 0.05]	[−0.12, 0.08]	[−0.17, −0.01]	[−0.38, −0.02]	[−0.63, −0.01]
	real	Proxy VAR	12	633	0.37	0.17	0.06	−0.94	−1.75	−2.45
					unbounded	unbounded	unbounded	unbounded	unbounded	unbounded
		Window impact	12	633	0.22*	0.15*	0.07	−0.12*	−0.26*	−0.34*
					[0.08, 0.40]	[0.03, 0.29]	[−0.06, 0.17]	[−0.22, −0.01]	[−0.48, −0.03]	[−0.74, −0.01]
r*-gap	none	Proxy VAR	12	633	−0.46*	−0.20	−0.10	−0.33	−0.72	−1.11
					[−2.02, 0.00]	[−0.98, 0.03]	[−0.52, 0.06]	[−2.16, 1.66]	[−3.76, 2.53]	[−5.14, 3.15]
		Window impact	12	633	−0.03	0.00	0.00	−0.13*	−0.30*	−0.45*
					[−0.15, 0.04]	[−0.11, 0.10]	[−0.10, 0.11]	[−0.22, −0.07]	[−0.48, −0.16]	[−0.77, −0.25]
	r*-gap	Proxy VAR	12	633	0.38	0.21	0.04	−0.82	−1.53	−2.09
					unbounded	unbounded	unbounded	unbounded	unbounded	unbounded
		Window impact	12	633	0.26*	0.16*	0.06	−0.14*	−0.30*	−0.41*
					[0.14, 0.39]	[0.06, 0.32]	[−0.04, 0.21]	[−0.24, −0.03]	[−0.52, −0.08]	[−0.76, −0.07]

Monthly VAR in the unemployment gap, core PCE inflation, the 2-year yield and, when included, the FCI (same step), sample as in the title, to September 2026, no trend, nothing dropped. Proxy VAR: Nakamura–Steinsson surprise as instrument; sets are exact 90% Anderson–Rubin sets, "unbounded" when the instrument cannot reject large effects; unit +1 pp on the monthly-average 2-year yield. Window impact: impact vector measured in the announcement window and scaled by the share of the month after the announcement; 90% bootstrap intervals; unit +1 pp 2-year in the window. The two units differ. * = 90% interval excludes zero.

Table 6: Monthly proxy VAR: first-stage strength and impact on the FCI

Start	2-year	FCI	F (3 lags)	F (6)	F (12)	Proxy impact on FCI	Window impact on FCI
1996	Nominal	none	6.5	5.9	5.2		
		nominal	3.5	2.9	1.9	0.10	0.99
	Real	none	7.8	6.3	5.2		
		real	4.9	4.0	2.4	0.16	0.99
	r*-gap	none	7.2	6.8	5.7		
		r*-gap	7.3	6.5	3.5	0.45	0.99
1973	Nominal	none	7.2	6.0	5.5		
		nominal	2.3	1.2	0.8	0.42	0.99
	Real	none	6.7	5.0	4.6		
		real	1.8	0.7	0.8	0.24	0.99
	r*-gap	none	6.6	5.6	5.0		
		r*-gap	1.6	0.7	0.9	0.63	0.99

F: robust first-stage statistic of the policy-variable residual on the instrument (instrument months only). The instrument exists from 1996, so the first stage uses the 1996–2026 instrument months in both samples. Impact on FCI: response of the FCI per +1 pp on the 2-year yield, estimated from monthly residuals (proxy VAR, 12 lags) and measured in the announcement window.

2 Robustness

Table 7: Robustness 1: sample (r*-gap 2-year yield and FCI) (12 lags)

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
Start 1973	none	12	633	−0.05	0.00	0.00	−0.23*	−0.50*	−0.78*
				[−0.22, 0.13]	[−0.19, 0.17]	[−0.19, 0.24]	[−0.36, −0.11]	[−0.79, −0.24]	[−1.25, −0.31]
	r*-gap	12	633	0.18	0.15	0.14	−0.27*	−0.62*	−0.94*
				[−0.04, 0.37]	[−0.01, 0.31]	[−0.04, 0.32]	[−0.44, −0.12]	[−0.94, −0.29]	[−1.46, −0.42]
2020 dropped	none	12	359	−0.10	−0.22	−0.26	−0.21	−0.33	−0.44
				[−0.33, 0.11]	[−0.68, 0.14]	[−0.87, 0.21]	[−0.39, 0.00]	[−0.76, 0.07]	[−1.17, 0.22]
	r*-gap	12	359	0.40*	0.45*	0.40*	−0.37*	−0.67*	−0.91*
				[0.22, 0.63]	[0.18, 0.72]	[0.05, 0.74]	[−0.57, −0.15]	[−1.11, −0.24]	[−1.62, −0.31]
2020 dropped, start 1973	none	12	623	0.18*	0.14	0.04	−0.29*	−0.64*	−0.96*
				[0.07, 0.29]	[0.00, 0.30]	[−0.14, 0.22]	[−0.41, −0.16]	[−0.89, −0.37]	[−1.35, −0.58]
	r*-gap	12	623	0.32*	0.24*	0.16	−0.32*	−0.74*	−1.11*
				[0.20, 0.43]	[0.10, 0.39]	[0.00, 0.32]	[−0.46, −0.16]	[−1.05, −0.41]	[−1.69, −0.56]
Ends 2019	none	12	288	−0.15	−0.27	−0.30	0.08	0.07	0.04
				[−0.36, 0.17]	[−0.67, 0.20]	[−0.86, 0.27]	[−0.06, 0.21]	[−0.11, 0.28]	[−0.21, 0.31]
	r*-gap	12	288	0.36*	0.31	0.18	−0.11	−0.21	−0.32
				[0.15, 0.54]	[−0.01, 0.58]	[−0.28, 0.58]	[−0.26, 0.05]	[−0.51, 0.07]	[−0.75, 0.08]
Ends 2019, start 1973	none	12	552	0.20*	0.18*	0.07	−0.22*	−0.51*	−0.83*
				[0.07, 0.35]	[0.01, 0.39]	[−0.14, 0.35]	[−0.35, −0.07]	[−0.82, −0.16]	[−1.36, −0.25]
	r*-gap	12	552	0.33*	0.25*	0.15	−0.23*	−0.54*	−0.90*
				[0.21, 0.47]	[0.10, 0.44]	[−0.03, 0.36]	[−0.41, −0.07]	[−0.94, −0.21]	[−1.53, −0.41]

Start 1973: monthly equations from 1973, with monthly series in place of daily data before 1990 (lib/model.py). 2020 dropped: March to December 2020 removed from the monthly and daily equations (2020 values still enter as lags). Ends 2019: all equations, monthly and daily. Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

Table 8: Robustness 2: the r* adjustment and trends (12 lags)

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36
Kiley r*	none	12	369	−0.54*	−0.31	−0.21	−0.09	−0.10	−0.
				[−0.93, −0.15]	[−0.64, 0.00]	[−0.58, 0.18]	[−0.27, 0.08]	[−0.46, 0.20]	[−0.71, 0.
	r*-gap (Kiley)	12	369	0.50*	0.45*	0.13	−0.27*	−0.21	−0.
				[0.09, 0.91]	[0.08, 0.96]	[−0.22, 0.58]	[−0.50, −0.04]	[−0.76, 0.34]	[−0.96, 0.
Kiley r*, start 1973	none	12	633	−0.11	−0.05	−0.02	−0.18*	−0.35*	−0.5
				[−0.25, 0.03]	[−0.21, 0.12]	[−0.18, 0.14]	[−0.31, −0.06]	[−0.61, −0.11]	[−0.90, −0.
	r*-gap (Kiley)	12	633	0.05	0.04	0.03	−0.20*	−0.37*	−0.4
				[−0.08, 0.25]	[−0.08, 0.18]	[−0.08, 0.17]	[−0.34, −0.06]	[−0.66, −0.12]	[−0.88, −0.
Real, spline trend	none	12	369	−0.66*	−0.34	−0.18	0.03	−0.06	−0.
				[−1.11, −0.27]	[−0.77, 0.08]	[−0.60, 0.30]	[−0.19, 0.21]	[−0.40, 0.30]	[−0.79, 0.
	real	12	369	0.25	0.39	0.28	−0.15	−0.28	−0.
				[−0.24, 0.71]	[−0.10, 0.88]	[−0.10, 0.71]	[−0.45, 0.07]	[−0.92, 0.20]	[−1.24, 0.
Real, spline trend, start 1973	none	12	633	−0.10	0.09	0.12	−0.10	−0.25*	−0.3
				[−0.31, 0.06]	[−0.22, 0.31]	[−0.17, 0.34]	[−0.19, 0.00]	[−0.44, −0.02]	[−0.64, −0.
	real	12	633	0.23*	0.27*	0.19*	−0.18*	−0.42*	−0.5
				[0.03, 0.42]	[0.06, 0.48]	[0.02, 0.36]	[−0.30, −0.03]	[−0.66, −0.16]	[−0.89, −0.

Kiley r*: 2-year minus expected inflation minus Kiley's r*, with Note A's index built with the same r*. Real, spline trend: real 2-year and real FCI with a natural cubic spline trend (knots at the sample ends and at NBER business-cycle peaks inside the sample) in every monthly equation, in place of any r* measure. Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

Table 9: Robustness 3: excess bond premium (r*-gap) (12 lags)

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
EBP added	none	12	369	−0.08	0.08	0.16	−0.27*	−0.52*	−0.77*
				[−0.42, 0.27]	[−0.39, 0.43]	[−0.43, 0.60]	[−0.41, −0.06]	[−0.87, −0.09]	[−1.36, −0.14]
	r*-gap	12	369	0.28	0.51*	0.40	−0.38*	−0.76*	−1.05*
				[−0.22, 0.81]	[0.03, 0.90]	[−0.08, 0.84]	[−0.57, −0.16]	[−1.16, −0.31]	[−1.71, −0.29]
EBP added, start 1973	none	12	633	−0.03	−0.01	0.01	−0.23*	−0.50*	−0.78*
				[−0.17, 0.15]	[−0.19, 0.16]	[−0.19, 0.18]	[−0.38, −0.11]	[−0.83, −0.23]	[−1.32, −0.28]
	r*-gap	12	633	0.07	0.04	0.04	−0.24*	−0.56*	−0.86*
				[−0.11, 0.25]	[−0.17, 0.24]	[−0.16, 0.24]	[−0.40, −0.08]	[−0.89, −0.22]	[−1.37, −0.37]

The excess bond premium is a third monthly variable, treated like unemployment and inflation (u, inflation, EBP system); responses shown for u and the price level. Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

A Results for 3, 6 and 12 lags

References cited in the monthly-VAR specification

- Bauer, M. D. and E. T. Swanson (2023), "A Reassessment of Monetary Policy Surprises and High-Frequency Identification," *NBER Macroeconomics Annual 2022*, Vol. 37, 87–155.
- Gertler, M. and P. Karadi (2015), "Monetary Policy Surprises, Credit Costs, and Economic Activity," *American Economic Journal: Macroeconomics* 7(1), 44–76.
- Lunsford, K. G. (2015), "Identifying Structural VARs with a Proxy Variable and a Test for a Weak Proxy," Federal Reserve Bank of Cleveland Working Paper 15-28.

Table 10: Baseline: 1996 to the present, all data in, no trend

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
Nominal	none	3	369	−0.60*	−0.43*	−0.31	0.11	0.08	0.04
				[−0.83, −0.28]	[−0.74, −0.06]	[−0.62, 0.06]	[−0.06, 0.29]	[−0.20, 0.39]	[−0.35, 0.48]
		6	369	−0.56*	−0.39*	−0.25	0.08	0.01	−0.11
				[−0.82, −0.27]	[−0.67, −0.08]	[−0.59, 0.09]	[−0.14, 0.28]	[−0.43, 0.35]	[−0.72, 0.40]
		12	369	−0.84*	−0.46*	−0.25	0.09	0.01	−0.13
				[−1.27, −0.39]	[−0.76, −0.13]	[−0.60, 0.10]	[−0.11, 0.34]	[−0.37, 0.45]	[−0.70, 0.51]
	nominal	3	369	0.03	0.20	0.20*	−0.11	−0.32	−0.46
				[−0.25, 0.35]	[−0.04, 0.50]	[0.03, 0.42]	[−0.33, 0.08]	[−0.73, 0.02]	[−1.01, 0.00]
		6	369	0.06	0.30	0.33*	−0.16	−0.45	−0.70*
				[−0.23, 0.34]	[−0.01, 0.60]	[0.07, 0.59]	[−0.45, 0.12]	[−1.00, 0.01]	[−1.43, −0.07]
		12	369	−0.01	0.30	0.38*	−0.15	−0.49*	−0.82*
				[−0.54, 0.45]	[−0.05, 0.60]	[0.05, 0.69]	[−0.41, 0.10]	[−0.94, −0.01]	[−1.43, −0.12]
Real	none	3	369	−0.36*	−0.33	−0.28	−0.05	−0.07	−0.09
				[−0.56, −0.04]	[−0.55, 0.04]	[−0.54, 0.06]	[−0.25, 0.11]	[−0.40, 0.17]	[−0.53, 0.27]
		6	369	−0.34	−0.29	−0.21	−0.01	−0.09	−0.20
				[−0.63, 0.01]	[−0.59, 0.08]	[−0.56, 0.13]	[−0.19, 0.18]	[−0.41, 0.21]	[−0.59, 0.23]
		12	369	−0.62*	−0.33	−0.20	−0.04	−0.11	−0.22
				[−0.96, −0.19]	[−0.64, 0.03]	[−0.52, 0.15]	[−0.24, 0.17]	[−0.46, 0.22]	[−0.76, 0.27]
	real	3	369	0.16	0.15	0.11	−0.21	−0.35	−0.42
				[−0.16, 0.43]	[−0.14, 0.40]	[−0.10, 0.23]	[−0.40, 0.02]	[−0.65, 0.02]	[−0.79, 0.02]
		6	369	0.18	0.29	0.25*	−0.21*	−0.47*	−0.67*
				[−0.07, 0.49]	[−0.01, 0.57]	[0.03, 0.47]	[−0.42, −0.02]	[−0.83, −0.10]	[−1.17, −0.08]
		12	369	0.13	0.33	0.31*	−0.25*	−0.53*	−0.73*
				[−0.30, 0.55]	[−0.03, 0.67]	[0.01, 0.63]	[−0.47, −0.03]	[−0.97, −0.07]	[−1.43, −0.07]
r*-gap	none	3	369	−0.17	−0.16	−0.13	−0.12	−0.15	−0.16
				[−0.43, 0.19]	[−0.50, 0.24]	[−0.44, 0.20]	[−0.26, 0.05]	[−0.42, 0.16]	[−0.54, 0.26]
		6	369	−0.08	−0.06	−0.01	−0.16	−0.35	−0.53
				[−0.42, 0.20]	[−0.49, 0.32]	[−0.48, 0.39]	[−0.36, 0.04]	[−0.74, 0.07]	[−1.07, 0.02]
		12	369	−0.28	−0.06	0.04	−0.22*	−0.44*	−0.68*
				[−0.91, 0.18]	[−0.68, 0.31]	[−0.56, 0.47]	[−0.40, −0.01]	[−0.82, −0.03]	[−1.34, −0.07]
	r*-gap	3	369	0.26	0.17	0.07	−0.20	−0.25	−0.27
				[−0.11, 0.54]	[−0.17, 0.41]	[−0.19, 0.21]	[−0.42, 0.00]	[−0.63, 0.09]	[−0.74, 0.19]
		6	369	0.33*	0.37*	0.28*	−0.32*	−0.61*	−0.84*
				[0.07, 0.54]	[0.07, 0.64]	[0.00, 0.49]	[−0.53, −0.11]	[−1.10, −0.21]	[−1.67, −0.24]
		12	369	0.33	0.50*	0.43*	−0.41*	−0.78*	−1.06*
				[−0.17, 0.67]	[0.12, 0.76]	[0.10, 0.70]	[−0.59, −0.19]	[−1.25, −0.35]	[−1.90, −0.32]

Nominal, real and r*-gap 2-year yield; the FCI is the matching index of Note A in each case. Per +1 pp on the 2-year yield.
* = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

Table 11: Robustness 1a: monthly equations start in 1973

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
r*-gap	none	3	642	0.01	0.05	0.05	−0.09	−0.15	−0.19
				[−0.10, 0.20]	[−0.10, 0.26]	[−0.10, 0.27]	[−0.23, 0.06]	[−0.48, 0.22]	[−0.71, 0.38]
		6	639	0.00	0.03	0.03	−0.12	−0.31	−0.52
				[−0.13, 0.12]	[−0.14, 0.18]	[−0.15, 0.17]	[−0.24, 0.02]	[−0.60, 0.02]	[−1.02, 0.01]
		12	633	−0.05	0.00	0.00	−0.23*	−0.50*	−0.78*
				[−0.22, 0.13]	[−0.19, 0.17]	[−0.19, 0.24]	[−0.36, −0.11]	[−0.79, −0.24]	[−1.25, −0.31]
	r*-gap	3	642	0.07	0.03	0.00	−0.08	−0.15	−0.23
				[−0.05, 0.18]	[−0.13, 0.17]	[−0.16, 0.16]	[−0.23, 0.07]	[−0.51, 0.18]	[−0.75, 0.28]
		6	639	0.10	0.08	0.07	−0.14	−0.35*	−0.59*
				[−0.06, 0.28]	[−0.11, 0.28]	[−0.13, 0.27]	[−0.29, 0.02]	[−0.67, −0.01]	[−1.11, −0.14]
		12	633	0.18	0.15	0.14	−0.27*	−0.62*	−0.94*
				[−0.04, 0.37]	[−0.01, 0.31]	[−0.04, 0.32]	[−0.44, −0.12]	[−0.94, −0.29]	[−1.46, −0.42]

r*-gap only. Pre-1990 months use monthly series in place of the daily data (see lib/model.py); shocks are the 252 announcements of 1996 on. Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

Table 12: Robustness 1b: 2020 dropped from all equations, start 1996

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
r*-gap	none	3	359	0.13	0.22	0.26	−0.18*	−0.28	−0.35
				[−0.01, 0.30]	[−0.01, 0.50]	[−0.02, 0.61]	[−0.38, −0.02]	[−0.65, 0.05]	[−0.81, 0.13]
		6	359	0.04	0.05	0.06	−0.15	−0.30	−0.44
				[−0.21, 0.25]	[−0.36, 0.41]	[−0.38, 0.52]	[−0.34, 0.05]	[−0.68, 0.08]	[−0.97, 0.16]
		12	359	−0.10	−0.22	−0.26	−0.21	−0.33	−0.44
				[−0.33, 0.11]	[−0.68, 0.14]	[−0.87, 0.21]	[−0.39, 0.00]	[−0.76, 0.07]	[−1.17, 0.22]
	r*-gap	3	359	0.36*	0.45*	0.41*	−0.22*	−0.34	−0.45
				[0.20, 0.53]	[0.21, 0.73]	[0.13, 0.74]	[−0.45, −0.02]	[−0.78, 0.04]	[−1.05, 0.07]
		6	359	0.37*	0.44*	0.44*	−0.29*	−0.56*	−0.78*
				[0.12, 0.55]	[0.07, 0.76]	[0.01, 0.89]	[−0.54, −0.06]	[−1.12, −0.04]	[−1.69, −0.01]
		12	359	0.40*	0.45*	0.40*	−0.37*	−0.67*	−0.91*
				[0.22, 0.63]	[0.18, 0.72]	[0.05, 0.74]	[−0.57, −0.15]	[−1.11, −0.24]	[−1.62, −0.31]

r*-gap only. March to December 2020 dropped from the monthly and the daily equations. Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

Table 13: Robustness 1c: 2020 dropped from all equations, start 1973

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
r*-gap	none	3	632	0.06	0.11	0.13	−0.11	−0.20	−0.27
				[−0.02, 0.14]	[−0.04, 0.24]	[−0.04, 0.30]	[−0.26, 0.03]	[−0.53, 0.13]	[−0.80, 0.22]
		6	629	0.11*	0.10	0.04	−0.17*	−0.39*	−0.62*
				[0.01, 0.22]	[−0.06, 0.24]	[−0.15, 0.22]	[−0.30, −0.03]	[−0.70, −0.05]	[−1.14, −0.12]
		12	623	0.18*	0.14	0.04	−0.29*	−0.64*	−0.96*
				[0.07, 0.29]	[0.00, 0.30]	[−0.14, 0.22]	[−0.41, −0.16]	[−0.89, −0.37]	[−1.35, −0.58]
	r*-gap	3	632	0.12*	0.10	0.08	−0.09	−0.24*	−0.44*
				[0.03, 0.22]	[−0.04, 0.25]	[−0.08, 0.24]	[−0.25, 0.00]	[−0.57, −0.01]	[−0.90, −0.05]
		6	629	0.27*	0.16*	0.07	−0.20*	−0.48*	−0.77*
				[0.16, 0.39]	[0.00, 0.32]	[−0.12, 0.26]	[−0.39, −0.07]	[−0.90, −0.18]	[−1.37, −0.30]
		12	623	0.32*	0.24*	0.16	−0.32*	−0.74*	−1.11*
				[0.20, 0.43]	[0.10, 0.39]	[0.00, 0.32]	[−0.46, −0.16]	[−1.05, −0.41]	[−1.69, −0.56]

r*-gap only. Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

Table 14: Robustness 1d: sample ends in December 2019, start 1996

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
r*-gap	none	3	288	−0.07	−0.13	−0.16	−0.03	0.00	0.03
				[−0.31, 0.15]	[−0.52, 0.23]	[−0.62, 0.29]	[−0.13, 0.06]	[−0.17, 0.17]	[−0.20, 0.26]
		6	288	−0.10	−0.19	−0.21	0.00	0.01	0.02
				[−0.43, 0.19]	[−0.79, 0.31]	[−0.96, 0.43]	[−0.12, 0.12]	[−0.21, 0.25]	[−0.27, 0.37]
		12	288	−0.15	−0.27	−0.30	0.08	0.07	0.04
				[−0.36, 0.17]	[−0.67, 0.20]	[−0.86, 0.27]	[−0.06, 0.21]	[−0.11, 0.28]	[−0.21, 0.31]
	r*-gap	3	288	0.30*	0.33	0.29	−0.14*	−0.14	−0.15
				[0.02, 0.56]	[−0.12, 0.68]	[−0.19, 0.65]	[−0.28, 0.00]	[−0.36, 0.08]	[−0.42, 0.14]
		6	288	0.28*	0.25	0.20	−0.14	−0.16	−0.18
				[0.04, 0.52]	[−0.18, 0.65]	[−0.37, 0.70]	[−0.28, 0.01]	[−0.38, 0.09]	[−0.48, 0.16]
		12	288	0.36*	0.31	0.18	−0.11	−0.21	−0.32
				[0.15, 0.54]	[−0.01, 0.58]	[−0.28, 0.58]	[−0.26, 0.05]	[−0.51, 0.07]	[−0.75, 0.08]

r*-gap only. All equations, monthly and daily. Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

Table 15: Robustness 1e: sample ends in December 2019, start 1973

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
r*-gap	none	3	561	0.03	0.05	0.06	−0.05	−0.06	−0.07
				[−0.06, 0.13]	[−0.09, 0.21]	[−0.10, 0.24]	[−0.19, 0.12]	[−0.45, 0.37]	[−0.67, 0.60]
		6	558	0.11*	0.09	0.04	−0.10	−0.27	−0.46
				[0.01, 0.24]	[−0.05, 0.28]	[−0.12, 0.23]	[−0.24, 0.05]	[−0.58, 0.07]	[−1.06, 0.10]
		12	552	0.20*	0.18*	0.07	−0.22*	−0.51*	−0.83*
				[0.07, 0.35]	[0.01, 0.39]	[−0.14, 0.35]	[−0.35, −0.07]	[−0.82, −0.16]	[−1.36, −0.25]
	r*-gap	3	561	0.08*	0.04	0.00	−0.01	−0.06	−0.19
				[0.00, 0.18]	[−0.11, 0.19]	[−0.18, 0.19]	[−0.14, 0.14]	[−0.33, 0.24]	[−0.60, 0.25]
		6	558	0.26*	0.13	0.02	−0.13	−0.34	−0.59*
				[0.15, 0.39]	[−0.04, 0.28]	[−0.15, 0.20]	[−0.28, 0.01]	[−0.67, 0.02]	[−1.12, −0.06]
		12	552	0.33*	0.25*	0.15	−0.23*	−0.54*	−0.90*
				[0.21, 0.47]	[0.10, 0.44]	[−0.03, 0.36]	[−0.41, −0.07]	[−0.94, −0.21]	[−1.53, −0.41]

r*-gap only. Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

Table 16: Robustness 2a: Kiley r*, start 1996

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
r*-gap (Kiley)	none	3	369	−0.33	−0.35	−0.30	−0.06	−0.04	−0.03
				[−0.60, 0.02]	[−0.63, 0.00]	[−0.62, 0.01]	[−0.27, 0.13]	[−0.34, 0.25]	[−0.40, 0.34]
		6	369	−0.29	−0.28	−0.22	−0.07	−0.12	−0.19
				[−0.55, 0.04]	[−0.63, 0.09]	[−0.62, 0.12]	[−0.28, 0.17]	[−0.54, 0.33]	[−0.77, 0.41]
		12	369	−0.54*	−0.31	−0.21	−0.09	−0.10	−0.13
				[−0.93, −0.15]	[−0.64, 0.00]	[−0.58, 0.18]	[−0.27, 0.08]	[−0.46, 0.20]	[−0.71, 0.31]
	r*-gap (Kiley)	3	369	0.25	0.07	−0.02	−0.11	−0.09	−0.08
				[−0.08, 0.62]	[−0.17, 0.31]	[−0.19, 0.14]	[−0.35, 0.13]	[−0.47, 0.31]	[−0.54, 0.40]
		6	369	0.44	0.30	0.12	−0.22	−0.31	−0.36
				[0.00, 0.72]	[−0.04, 0.63]	[−0.20, 0.40]	[−0.47, 0.05]	[−0.80, 0.24]	[−1.07, 0.44]
		12	369	0.50*	0.45*	0.13	−0.27*	−0.21	−0.03
				[0.09, 0.91]	[0.08, 0.96]	[−0.22, 0.58]	[−0.50, −0.04]	[−0.76, 0.34]	[−0.96, 0.81]

2-year minus expected inflation minus Kiley's r*; the FCI is Note A's index built with Kiley's r* (same adjustment in rate and FCI). Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

Table 17: Robustness 2b: Kiley r*, start 1973

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
r*-gap (Kiley)	none	3	642	−0.02	0.00	0.01	−0.04	−0.02	0.01
				[−0.16, 0.09]	[−0.15, 0.18]	[−0.13, 0.18]	[−0.17, 0.10]	[−0.33, 0.29]	[−0.46, 0.52]
		6	639	−0.04	−0.01	0.00	−0.07	−0.17	−0.28
				[−0.15, 0.07]	[−0.16, 0.16]	[−0.16, 0.15]	[−0.18, 0.04]	[−0.42, 0.08]	[−0.70, 0.15]
		12	633	−0.11	−0.05	−0.02	−0.18*	−0.35*	−0.50*
				[−0.25, 0.03]	[−0.21, 0.12]	[−0.18, 0.14]	[−0.31, −0.06]	[−0.61, −0.11]	[−0.90, −0.10]
	r*-gap (Kiley)	3	642	−0.02	−0.05	−0.05	−0.04	−0.06	−0.07
				[−0.11, 0.11]	[−0.15, 0.08]	[−0.15, 0.05]	[−0.17, 0.08]	[−0.29, 0.15]	[−0.38, 0.23]
		6	639	−0.01	−0.01	0.00	−0.08	−0.18	−0.27
				[−0.12, 0.12]	[−0.14, 0.13]	[−0.13, 0.14]	[−0.20, 0.04]	[−0.42, 0.06]	[−0.63, 0.08]
		12	633	0.05	0.04	0.03	−0.20*	−0.37*	−0.48*
				[−0.08, 0.25]	[−0.08, 0.18]	[−0.08, 0.17]	[−0.34, −0.06]	[−0.66, −0.12]	[−0.88, −0.15]

As 2a. Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

Table 18: Robustness 2c: real 2-year and real FCI with a cubic spline trend, start 1996

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
Real	none	3	369	−0.46*	−0.40*	−0.32*	−0.07	−0.15	−0.22
				[−0.69, −0.15]	[−0.67, −0.10]	[−0.62, −0.06]	[−0.21, 0.10]	[−0.40, 0.15]	[−0.54, 0.18]
		6	369	−0.46*	−0.34	−0.22	−0.03	−0.22	−0.42
				[−0.72, −0.13]	[−0.71, 0.00]	[−0.58, 0.12]	[−0.24, 0.15]	[−0.62, 0.15]	[−1.00, 0.11]
		12	369	−0.66*	−0.34	−0.18	0.03	−0.06	−0.22
				[−1.11, −0.27]	[−0.77, 0.08]	[−0.60, 0.30]	[−0.19, 0.21]	[−0.40, 0.30]	[−0.79, 0.29]
	real	3	369	0.04	0.05	0.05	−0.18	−0.31	−0.39
				[−0.28, 0.42]	[−0.27, 0.34]	[−0.22, 0.20]	[−0.47, 0.02]	[−0.79, 0.06]	[−0.93, 0.07]
		6	369	0.16	0.29	0.25*	−0.22	−0.52*	−0.72*
				[−0.17, 0.63]	[−0.06, 0.71]	[0.02, 0.51]	[−0.43, 0.01]	[−0.99, −0.05]	[−1.34, −0.05]
		12	369	0.25	0.39	0.28	−0.15	−0.28	−0.30
				[−0.24, 0.71]	[−0.10, 0.88]	[−0.10, 0.71]	[−0.45, 0.07]	[−0.92, 0.20]	[−1.24, 0.40]

No r* measure: the trend is a natural cubic spline in time (knots at the sample ends and at the NBER business-cycle peaks inside the sample), estimated in every monthly equation. Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

Table 19: Robustness 2d: real 2-year and real FCI with a cubic spline trend, start 1973

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
Real	none	3	642	0.01	0.04	0.04	−0.11*	−0.22*	−0.29*
				[−0.17, 0.19]	[−0.18, 0.24]	[−0.15, 0.19]	[−0.23, −0.01]	[−0.44, −0.03]	[−0.60, −0.05]
		6	639	0.02	0.07	0.06	−0.11*	−0.32*	−0.48*
				[−0.14, 0.14]	[−0.14, 0.24]	[−0.13, 0.20]	[−0.23, −0.01]	[−0.55, −0.04]	[−0.81, −0.13]
		12	633	−0.10	0.09	0.12	−0.10	−0.25*	−0.39*
				[−0.31, 0.06]	[−0.22, 0.31]	[−0.17, 0.34]	[−0.19, 0.00]	[−0.44, −0.02]	[−0.64, −0.06]
	real	3	642	0.26*	0.23*	0.15*	−0.18*	−0.33*	−0.41*
				[0.06, 0.43]	[0.06, 0.37]	[0.04, 0.24]	[−0.28, −0.05]	[−0.52, −0.07]	[−0.65, −0.08]
		6	639	0.27*	0.29*	0.19*	−0.18*	−0.46*	−0.65*
				[0.06, 0.48]	[0.08, 0.49]	[0.06, 0.34]	[−0.32, −0.05]	[−0.73, −0.16]	[−1.00, −0.25]
		12	633	0.23*	0.27*	0.19*	−0.18*	−0.42*	−0.57*
				[0.03, 0.42]	[0.06, 0.48]	[0.02, 0.36]	[−0.30, −0.03]	[−0.66, −0.16]	[−0.89, −0.19]

As 2c; knots at the NBER peaks of 1973, 1980, 1981, 1990, 2001, 2007 and 2020. Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

Table 20: Robustness 3a: excess bond premium as a third monthly variable, start 1996

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
r*-gap	none	3	369	−0.11	−0.14	−0.13	−0.14	−0.17	−0.18
				[−0.43, 0.28]	[−0.55, 0.27]	[−0.54, 0.20]	[−0.35, 0.03]	[−0.55, 0.12]	[−0.67, 0.24]
		6	369	0.02	−0.05	−0.08	−0.19	−0.36	−0.52
				[−0.33, 0.36]	[−0.46, 0.34]	[−0.53, 0.32]	[−0.38, 0.06]	[−0.74, 0.06]	[−1.09, 0.12]
		12	369	−0.08	0.08	0.16	−0.27*	−0.52*	−0.77*
				[−0.42, 0.27]	[−0.39, 0.43]	[−0.43, 0.60]	[−0.41, −0.06]	[−0.87, −0.09]	[−1.36, −0.14]
	r*-gap	3	369	0.20	0.05	−0.11	−0.18	−0.19	−0.17
				[−0.13, 0.51]	[−0.37, 0.44]	[−0.59, 0.28]	[−0.37, 0.03]	[−0.61, 0.21]	[−0.71, 0.38]
		6	369	0.31	0.23	0.05	−0.29*	−0.54*	−0.74*
				[−0.04, 0.66]	[−0.13, 0.67]	[−0.36, 0.51]	[−0.52, −0.08]	[−0.96, −0.13]	[−1.44, −0.08]
		12	369	0.28	0.51*	0.40	−0.38*	−0.76*	−1.05*
				[−0.22, 0.81]	[0.03, 0.90]	[−0.08, 0.84]	[−0.57, −0.16]	[−1.16, −0.31]	[−1.71, −0.29]

r*-gap. Monthly system u, inflation, EBP (EBP treated like u and inflation; responses shown for u and the price level). Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

Table 21: Robustness 3b: excess bond premium as a third monthly variable, start 1973

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
r*-gap	none	3	642	0.01	0.04	0.04	−0.09	−0.15	−0.19
				[−0.10, 0.15]	[−0.10, 0.21]	[−0.09, 0.19]	[−0.25, 0.06]	[−0.52, 0.23]	[−0.72, 0.42]
		6	639	0.01	0.03	0.01	−0.12*	−0.31*	−0.51*
				[−0.10, 0.14]	[−0.14, 0.19]	[−0.19, 0.16]	[−0.26, −0.02]	[−0.61, −0.04]	[−1.06, −0.09]
		12	633	−0.03	−0.01	0.01	−0.23*	−0.50*	−0.78*
				[−0.17, 0.15]	[−0.19, 0.16]	[−0.19, 0.18]	[−0.38, −0.11]	[−0.83, −0.23]	[−1.32, −0.28]
	r*-gap	3	642	0.03	0.02	0.00	−0.09	−0.19	−0.28
				[−0.09, 0.14]	[−0.13, 0.16]	[−0.16, 0.14]	[−0.22, 0.08]	[−0.51, 0.18]	[−0.75, 0.29]
		6	639	0.04	0.03	0.01	−0.12	−0.31	−0.55
				[−0.09, 0.19]	[−0.15, 0.18]	[−0.18, 0.15]	[−0.26, 0.03]	[−0.62, 0.02]	[−1.04, 0.04]
		12	633	0.07	0.04	0.04	−0.24*	−0.56*	−0.86*
				[−0.11, 0.25]	[−0.17, 0.24]	[−0.16, 0.24]	[−0.40, −0.08]	[−0.89, −0.22]	[−1.37, −0.37]

As 3a. Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

Table 22: Monthly VAR benchmark, 1996 sample (3, 6 and 12 lags)

					Unemployment gap (pp)			Price level (%)		
2-year	FCI	Identification	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
Nominal	none	Proxy VAR	3	369	−0.68*	−0.54*	−0.44*	0.03	−0.01	−0.05
					[−1.12, −0.10]	[−0.78, −0.23]	[−0.59, −0.24]	[−1.05, 0.91]	[−1.11, 0.89]	[−1.15, 0.84]
			6	369	−0.58	−0.45	−0.33	−0.02	−0.13	−0.28
					[−1.13, 0.24]	[−0.91, 0.23]	[−0.68, 0.19]	[−1.42, 0.98]	[−1.95, 1.16]	[−2.23, 1.10]
			12	369	−1.27*	−0.84*	−0.45	0.07	−0.07	−0.32
					[−1.93, −0.68]	[−1.42, −0.25]	[−0.92, 0.06]	[−1.31, 1.25]	[−2.05, 1.59]	[−2.58, 1.56]
		Window impact	3	369	−0.35*	−0.27*	−0.21	0.07	0.05	0.03
					[−0.54, −0.17]	[−0.52, −0.04]	[−0.53, 0.02]	[−0.03, 0.19]	[−0.10, 0.26]	[−0.17, 0.36]
			6	369	−0.30*	−0.23*	−0.17*	0.04	0.00	−0.06
					[−0.49, −0.17]	[−0.44, −0.08]	[−0.41, 0.00]	[−0.06, 0.17]	[−0.16, 0.19]	[−0.30, 0.24]
			12	369	−0.57*	−0.39*	−0.21	0.07	0.02	−0.08
					[−0.88, −0.33]	[−0.74, −0.17]	[−0.57, 0.05]	[−0.07, 0.21]	[−0.26, 0.33]	[−0.48, 0.33]
	nominal	Proxy VAR	3	369	−0.58	−0.68	−0.59	−0.10	−0.04	−0.01
					[−1.84, 2.02]	[−2.69, 0.45]	[−2.56, 0.13]	[−2.90, 1.39]	[−2.44, 1.85]	[−2.16, 2.19]
			6	369	−0.38	−0.53	−0.47	−0.31	−0.40	−0.47
					[−2.11, 12.57]	[−4.61, 3.48]	[−5.01, 1.23]	[−16.09, 1.53]	[−17.38, 2.36]	[−15.83, 2.82]
			12	369	−0.87	−0.80	−0.51	−0.36	−0.47	−0.52
					unbounded	unbounded	unbounded	unbounded	unbounded	unbounded
		Window impact	3	369	−0.07	−0.01	−0.01	−0.04	−0.13	−0.19
					[−0.23, 0.12]	[−0.23, 0.24]	[−0.26, 0.26]	[−0.16, 0.08]	[−0.35, 0.07]	[−0.49, 0.10]
			6	369	0.00	0.05	0.02	−0.08	−0.21	−0.31
					[−0.21, 0.24]	[−0.17, 0.31]	[−0.20, 0.31]	[−0.25, 0.08]	[−0.51, 0.07]	[−0.76, 0.09]
			12	369	−0.16	−0.10	−0.05	−0.05	−0.20	−0.33
					[−0.47, 0.12]	[−0.41, 0.23]	[−0.39, 0.27]	[−0.20, 0.08]	[−0.43, 0.06]	[−0.65, 0.09]
Real	none	Proxy VAR	3	369	−0.53*	−0.45*	−0.38*	−0.06	−0.09	−0.11
					[−1.14, −0.10]	[−0.79, −0.20]	[−0.61, −0.21]	[−0.99, 0.81]	[−1.05, 0.81]	[−1.06, 0.78]
			6	369	−0.54	−0.40	−0.26	−0.14	−0.30	−0.47
					[−1.36, 0.10]	[−1.04, 0.21]	[−0.73, 0.23]	[−1.47, 0.95]	[−2.05, 1.13]	[−2.35, 1.05]
			12	369	−0.89*	−0.49	−0.16	−0.31	−0.58	−0.83
					[−1.77, −0.31]	[−1.23, 0.24]	[−0.74, 0.54]	[−2.00, 0.87]	[−3.03, 1.05]	[−3.68, 0.97]
		Window impact	3	369	−0.20*	−0.19*	−0.16*	−0.03	−0.04	−0.05
					[−0.32, −0.05]	[−0.33, −0.03]	[−0.33, −0.01]	[−0.13, 0.06]	[−0.21, 0.11]	[−0.27, 0.18]
			6	369	−0.22*	−0.18*	−0.13	−0.01	−0.06	−0.13
					[−0.42, −0.06]	[−0.42, −0.01]	[−0.38, 0.07]	[−0.13, 0.14]	[−0.27, 0.21]	[−0.41, 0.26]
			12	369	−0.39*	−0.25*	−0.12	−0.02	−0.08	−0.15
					[−0.69, −0.21]	[−0.60, −0.02]	[−0.45, 0.16]	[−0.14, 0.10]	[−0.32, 0.15]	[−0.56, 0.21]
	real	Proxy VAR	3	369	−0.42	−0.58	−0.53	−0.16	−0.08	−0.02
					[−1.33, 0.90]	[−1.45, 0.28]	[−1.22, 0.08]	[−1.71, 0.92]	[−1.59, 1.09]	[−1.50, 1.18]
			6	369	−0.30	−0.49	−0.46	−0.40	−0.51	−0.57
					[−1.42, 2.38]	[−1.67, 1.36]	[−1.32, 0.62]	[−3.52, 0.88]	[−4.38, 1.20]	[−4.47, 1.26]
			12	369	−0.38	−0.46	−0.34	−0.72	−0.87	−0.82
					unbounded	unbounded	unbounded	unbounded	unbounded	unbounded
		Window impact	3	369	0.05	0.02	−0.01	−0.11	−0.17	−0.20
					[−0.12, 0.26]	[−0.16, 0.28]	[−0.19, 0.22]	[−0.23, 0.01]	[−0.36, 0.01]	[−0.42, 0.03]
			6	369	0.06	0.07	0.01	−0.13	−0.26	−0.35
					[−0.15, 0.28]	[−0.16, 0.28]	[−0.22, 0.24]	[−0.27, 0.02]	[−0.49, 0.00]	[−0.67, 0.01]
			12	369	−0.02	0.00	0.01	−0.13	−0.26	−0.33
					[−0.32, 0.26]	[−0.32, 0.27]	[−0.33, 0.35]	[−0.32, 0.04]	[−0.60, 0.06]	[−0.78, 0.08]
r*-gap	none	Proxy VAR	3	369	−0.36	−0.27*	−0.20*	−0.13	−0.17	−0.19
					[−1.17, 0.10]	[−0.70, −0.01]	[−0.44, −0.05]	[−1.07, 0.79]	[−1.13, 0.78]	[−1.15, 0.75]
			6	369	−0.27	−0.14	−0.02	−0.26	−0.51	−0.73
					[−1.25, 0.29]	[−0.85, 0.34]	[−0.50, 0.33]	[−1.34, 0.86]	[−1.92, 0.91]	[−2.20, 0.73]
			12	369	−0.47	−0.07	0.15	−0.51	−0.89	−1.20
					[−1.52, 0.08]	[−0.86, 0.52]	[−0.37, 0.63]	[−1.90, 0.56]	[−2.88, 0.55]	[−3.46, 0.34]
		Window impact	3	369	−0.09	−0.09	−0.07	−0.06	−0.08	−0.09
					[−0.28, 0.09]	[−0.34, 0.20]	[−0.34, 0.20]	[−0.18, 0.04]	[−0.26, 0.11]	[−0.35, 0.17]
			6	369	−0.05	−0.03	0.01	−0.10	−0.20	−0.30
					[−0.24, 0.15]	[−0.28, 0.23]	[−0.26, 0.29]	[−0.23, 0.02]	[−0.47, 0.03]	[−0.69, 0.03]
			12	369	−0.15	−0.02	0.05	−0.12*	−0.24*	−0.34*
					[−0.42, 0.11]	[−0.26, 0.22]	[−0.14, 0.29]	[−0.25, −0.01]	[−0.49, −0.01]	[−0.68, −0.02]
	r*-gap	Proxy VAR	3	369	−0.20	−0.33	−0.36	−0.23	−0.24	−0.24
					[−0.97, 0.75]	[−0.95, 0.39]	[−0.83, 0.18]	[−1.26, 0.60]	[−1.29, 0.60]	[−1.29, 0.60]
			6	369	0.01	−0.14	−0.22	−0.53	−0.80	−0.97
					[−1.00, 1.64]	[−0.96, 1.11]	[−0.74, 0.55]	[−2.11, 0.40]	[−2.82, 0.39]	[−3.01, 0.24]
			12	369	−0.06	−0.06	−0.10	−0.89	−1.30	−1.43
					[−1.82, 5.22]	[−1.57, 3.84]	[−1.18, 2.00]	[−6.80, 0.28]	[−9.27, 0.41]	[−9.64, 0.47]
		Window impact	3	369	0.13	0.06	0.00	−0.11	−0.13	−0.14
					[−0.05, 0.34]	[−0.17, 0.30]	[−0.24, 0.22]	[−0.24, 0.02]	[−0.35, 0.10]	[−0.42, 0.16]
			6	369	0.21	0.15	0.04	−0.20*	−0.34*	−0.43*
					[−0.01, 0.45]	[−0.11, 0.47]	[−0.19, 0.35]	[−0.38, −0.07]	[−0.64, −0.11]	[−0.85, −0.11]

Table 23: Monthly VAR benchmark, 1973 sample (3, 6 and 12 lags)

					Unemployment gap (pp)			Price level (%)		
2-year	FCI	Identification	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
Nominal	none	Proxy VAR	3	642	−0.27	−0.13	−0.04	0.11	0.38	0.69
					[−0.69, 0.03]	[−0.38, 0.05]	[−0.20, 0.06]	[−1.51, 1.43]	[−1.72, 2.10]	[−1.68, 2.62]
			6	639	−0.36*	−0.18	−0.07	0.09	0.20	0.30
					[−0.85, −0.05]	[−0.48, 0.01]	[−0.28, 0.05]	[−1.67, 1.36]	[−2.64, 2.25]	[−3.38, 2.96]
			12	633	−0.57*	−0.30*	−0.11	0.05	0.09	0.09
					[−1.31, −0.16]	[−0.69, −0.09]	[−0.34, 0.03]	[−1.69, 1.59]	[−2.76, 2.62]	[−3.69, 3.45]
		Window impact	3	642	−0.05	−0.01	0.01	0.10*	0.24*	0.38*
					[−0.10, 0.01]	[−0.08, 0.05]	[−0.06, 0.08]	[0.02, 0.17]	[0.09, 0.36]	[0.17, 0.57]
			6	639	−0.09*	−0.03	0.00	0.09*	0.16*	0.23
					[−0.17, −0.03]	[−0.10, 0.06]	[−0.08, 0.10]	[0.01, 0.16]	[0.01, 0.32]	[−0.02, 0.46]
			12	633	−0.16*	−0.08	−0.02	0.05	0.09	0.12
					[−0.26, −0.05]	[−0.18, 0.00]	[−0.11, 0.07]	[−0.04, 0.14]	[−0.13, 0.27]	[−0.24, 0.40]
	nominal	Proxy VAR	3	642	−0.13	−0.20	−0.18	−0.12	−0.13	−0.10
					unbounded	unbounded	unbounded	unbounded	unbounded	unbounded
			6	639	−0.09	−0.14	−0.14	−0.14	−0.35	−0.58
					unbounded	unbounded	unbounded	unbounded	unbounded	unbounded
			12	633	0.33	0.21	0.15	−0.50	−0.93	−1.34
					unbounded	unbounded	unbounded	unbounded	unbounded	unbounded
		Window impact	3	642	0.05	0.05	0.05	0.08	0.21*	0.35*
					[0.00, 0.11]	[0.00, 0.12]	[−0.01, 0.12]	[0.00, 0.15]	[0.07, 0.34]	[0.14, 0.53]
			6	639	0.07	0.07*	0.07*	0.07	0.15	0.24
					[−0.01, 0.18]	[0.01, 0.16]	[0.01, 0.15]	[−0.02, 0.16]	[−0.02, 0.32]	[−0.03, 0.47]
			12	633	0.15*	0.11*	0.06	0.04	0.04	0.06
					[0.01, 0.34]	[0.00, 0.30]	[−0.02, 0.21]	[−0.08, 0.16]	[−0.22, 0.29]	[−0.33, 0.40]
Real	none	Proxy VAR	3	642	−0.34	−0.19	−0.10	−0.17	−0.15	−0.07
					[−1.15, 0.00]	[−0.69, 0.02]	[−0.41, 0.03]	[−2.63, 1.55]	[−3.54, 2.22]	[−3.90, 2.61]
			6	639	−0.41*	−0.23*	−0.13	−0.10	−0.21	−0.32
					[−1.62, −0.04]	[−0.98, 0.00]	[−0.62, 0.03]	[−2.49, 1.76]	[−4.08, 2.85]	[−5.34, 3.68]
			12	633	−0.56*	−0.30*	−0.14	−0.25	−0.51	−0.75
					[−2.14, −0.11]	[−1.14, −0.06]	[−0.61, 0.02]	[−2.58, 1.81]	[−4.36, 2.88]	[−5.84, 3.70]
		Window impact	3	642	−0.03	−0.01	0.00	0.00	0.04	0.09
					[−0.09, 0.04]	[−0.11, 0.09]	[−0.10, 0.11]	[−0.08, 0.08]	[−0.15, 0.22]	[−0.22, 0.36]
			6	639	−0.05	−0.02	−0.01	−0.03	−0.08	−0.13
					[−0.12, 0.03]	[−0.10, 0.08]	[−0.10, 0.10]	[−0.10, 0.06]	[−0.24, 0.11]	[−0.40, 0.17]
			12	633	−0.10	−0.05	−0.02	−0.10*	−0.21*	−0.30*
					[−0.21, 0.00]	[−0.16, 0.05]	[−0.12, 0.08]	[−0.17, −0.01]	[−0.38, −0.02]	[−0.63, −0.01]
	real	Proxy VAR	3	642	−0.14	−0.21	−0.21	−0.40	−0.78	−1.06
					unbounded	unbounded	unbounded	unbounded	unbounded	unbounded
			6	639	−0.03	−0.15	−0.19	−0.38	−0.91	−1.45
					unbounded	unbounded	unbounded	unbounded	unbounded	unbounded
			12	633	0.37	0.17	0.06	−0.94	−1.75	−2.45
					unbounded	unbounded	unbounded	unbounded	unbounded	unbounded
		Window impact	3	642	0.07*	0.06	0.05	0.00	0.07	0.15
					[0.02, 0.14]	[−0.01, 0.15]	[−0.02, 0.14]	[−0.08, 0.08]	[−0.14, 0.19]	[−0.15, 0.36]
			6	639	0.11*	0.08*	0.06	−0.05	−0.08	−0.10
					[0.02, 0.24]	[0.00, 0.19]	[−0.02, 0.16]	[−0.14, 0.05]	[−0.28, 0.12]	[−0.44, 0.20]
			12	633	0.22*	0.15*	0.07	−0.12*	−0.26*	−0.34*
					[0.08, 0.40]	[0.03, 0.29]	[−0.06, 0.17]	[−0.22, −0.01]	[−0.48, −0.03]	[−0.74, −0.01]
r*-gap	none	Proxy VAR	3	642	−0.28	−0.12	−0.04	−0.33	−0.49	−0.58
					[−1.17, 0.07]	[−0.66, 0.09]	[−0.37, 0.09]	[−2.74, 1.49]	[−3.82, 2.02]	[−4.32, 2.24]
			6	639	−0.32	−0.15	−0.08	−0.19	−0.45	−0.72
					[−1.42, 0.05]	[−0.81, 0.07]	[−0.51, 0.09]	[−2.10, 1.67]	[−3.54, 2.58]	[−4.71, 3.20]
			12	633	−0.46*	−0.20	−0.10	−0.33	−0.72	−1.11
					[−2.02, 0.00]	[−0.98, 0.03]	[−0.52, 0.06]	[−2.16, 1.66]	[−3.76, 2.53]	[−5.14, 3.15]
		Window impact	3	642	0.01	0.03	0.03	−0.05	−0.09	−0.11
					[−0.07, 0.08]	[−0.07, 0.11]	[−0.07, 0.12]	[−0.13, 0.05]	[−0.30, 0.15]	[−0.45, 0.28]
			6	639	0.00	0.02	0.02	−0.08	−0.19*	−0.32*
					[−0.08, 0.09]	[−0.09, 0.14]	[−0.10, 0.13]	[−0.15, 0.00]	[−0.38, −0.02]	[−0.64, −0.03]
			12	633	−0.03	0.00	0.00	−0.13*	−0.30*	−0.45*
					[−0.15, 0.04]	[−0.11, 0.10]	[−0.10, 0.11]	[−0.22, −0.07]	[−0.48, −0.16]	[−0.77, −0.25]
	r*-gap	Proxy VAR	3	642	−0.15	−0.13	−0.11	−0.40	−0.69	−0.94
					unbounded	unbounded	unbounded	unbounded	unbounded	unbounded
			6	639	−0.01	−0.07	−0.12	−0.43	−0.86	−1.30
					unbounded	unbounded	unbounded	unbounded	unbounded	unbounded
			12	633	0.38	0.21	0.04	−0.82	−1.53	−2.09
					unbounded	unbounded	unbounded	unbounded	unbounded	unbounded
		Window impact	3	642	0.10*	0.09*	0.08*	−0.02	0.01	0.04
					[0.04, 0.17]	[0.02, 0.17]	[0.00, 0.15]	[−0.10, 0.07]	[−0.17, 0.17]	[−0.25, 0.31]
			6	639	0.18*	0.12*	0.07	−0.08	−0.16	−0.24
					[0.08, 0.27]	[0.04, 0.23]	[0.00, 0.17]	[−0.19, 0.03]	[−0.41, 0.04]	[−0.64, 0.07]

- Mertens, K. and M. O. Ravn (2013), "The Dynamic Effects of Personal and Corporate Income Tax Changes in the United States," *American Economic Review* 103(4), 1212–1247.
- Montiel Olea, J. L., J. H. Stock and M. W. Watson (2021), "Inference in Structural Vector Autoregressions Identified with an External Instrument," *Journal of Econometrics* 225(1), 74–87.
- Nakamura, E. and J. Steinsson (2018), "High-Frequency Identification of Monetary Non-Neutrality: The Information Effect," *Quarterly Journal of Economics* 133(3), 1283–1330.
- Plagborg-Møller, M. and C. K. Wolf (2021), "Local Projections and VARs Estimate the Same Impulse Responses," *Econometrica* 89(2), 955–980.
- Stock, J. H. and M. W. Watson (2018), "Identification and Estimation of Dynamic Causal Effects in Macroeconomics Using External Instruments," *Economic Journal* 128(610), 917–948.